

**ASSESSING THE EFFECTIVENESS AND ACCEPTABILITY OF
PHOTOVOLTAIC ARRAY MONITORING SYSTEM AT CEBU
TECHNOLOGICAL UNIVERSITY-TUBURAN CAMPUS**

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Chapter 1

THE PROBLEM AND ITS SCOPE

INTRODUCTION

Rationale of the Study

The growing demand for energy like electricity, and the urgent need to address climate change highlight the importance of finding reliable, affordable, and sustainable energy sources. One of the few things that addresses this concerning problem is by the use of solar panels. As a clean, sustainable, and renewable substitute for fossil fuels, solar energy helps lower greenhouse gas emissions and slow down environmental deterioration (IEA, 2023). With technological advancements and decreasing installation costs, solar power like photovoltaic solar improves community resilience by providing dependable electricity even in the event of natural disasters and power outages.

In the recent years, the rapid expansion of solar photovoltaic (PV) technology has major significant effect on the environment and the economy. Globally, the use of solar photovoltaic (PV) systems has expanded rapidly, but many nations still struggle to ensure their efficiency and sustainability. In the United States, issues such as the intermittency of solar power and the lack of thorough recycling regulations for decommissioned panels (Paiano, 2022). The world's largest producer of solar panels is the country of China, and it faces issues with environmental pollution and material sustainability because of the extraction and processing of silicon and rare metals used in PV manufacturing (Zhao et al., 2023). The Global South, which is made up of economically disadvantaged parts of the world, has a number of difficulties,

including limited access to clean water, power, industrialization, and food security. As a plentiful and sustainable resource, solar energy has enormous potential to solve these issues (Ukoba et al., 2024).

Meanwhile, in the Philippines, the adoption of solar panels has continued to become common; however, high installation costs, policy inconsistencies, and grid infrastructure limitations persist as significant barriers. According to the Department of Energy (DOE, 2023), even though the nation has a lot of solar energy—about 5.1 kWh/m²/day—renewable energy still makes up less than 30% of the power mix. Many solar installations, particularly in rural or off-grid areas, still face ineffective monitoring and maintenance that reduces system performance and shortens lifespan (Rosauero et al., 2025). Furthermore, maintenance issues in rural or isolated locations, where there are few skilled specialists, result in shorter system lifespans and inefficiencies.

Solar energy solutions in Cebu City is actively pursuing to address its escalating electricity costs and frequent power outages. In an effort to save the city's yearly electricity costs by as much as ₱50 million, mayor-elect Nestor Archival has started the installation of a 300-kilowatt solar power system at the South Road Properties (Philippine Star, 2025). The city's commitment to smart, resilient, and sustainable energy solutions is demonstrated by the integration of solar energy into public infrastructure and IoT-enabled management. One of the schools in Tuburan, Cebu specifically Cebu Technological University – Tuburan Campus has utilized solar energy to help helps fossil fuel use and lower electricity bills. The solar photovoltaic array powers the campus water system, which supplies water to the entire

university, including classrooms, laboratories, kitchens, and irrigation. This system supports environmental sustainability and ensures a steady water supply in an area prone to power interruptions.

There are various research studies that aimed to preserve environmental sustainability and conserve energy that utilized innovative solutions like the Internet of Things(IoT) integrated in the use of solar panels. However, several challenges continue to limit its efficiency, reliability, and long-term sustainability. By enabling real-time data collection, remote monitoring, defect detection, and predictive maintenance, the incorporation of Internet of Things (IoT) technology has encouraging prospects to overcome these constraints. Thus, there is insufficient evaluation of system reliability and accuracy under extreme weather conditions, partial shading, and tropical climates. Therefore, developing a simple, cost-effective IoT monitoring system for PV arrays is important, particularly in real-world conditions. It truly enhances energy efficiency, reduce operational costs, and support the sustainable expansion of solar energy.

The proposed IoT monitoring system focuses only on monitoring the water level supply in the deep wheel and displaying key information from the solar photovoltaic array. It shows important data such as voltage, current (ampere), temperature, and energy consumption per hour and per day. Users can view this information remotely through an easy-to-use dashboard in an application and in person. The system provides real-time monitoring and data display. Using affordable components like ESP32 microcontrollers, a flow meter, and low-power sensors makes the system practical and suitable for the campus. This study supported the UN Sustainable Development Goals by

improving access to clean and reliable energy (SDG 7) and strengthening innovation and infrastructure through the implementation of an IoT-based solar monitoring system (SDG 9). It also contributes to climate action (SDG 13) by enhancing solar energy efficiency, thereby reducing greenhouse gas emissions and reliance on fossil fuels, which is vital in disaster-prone areas like Cebu.

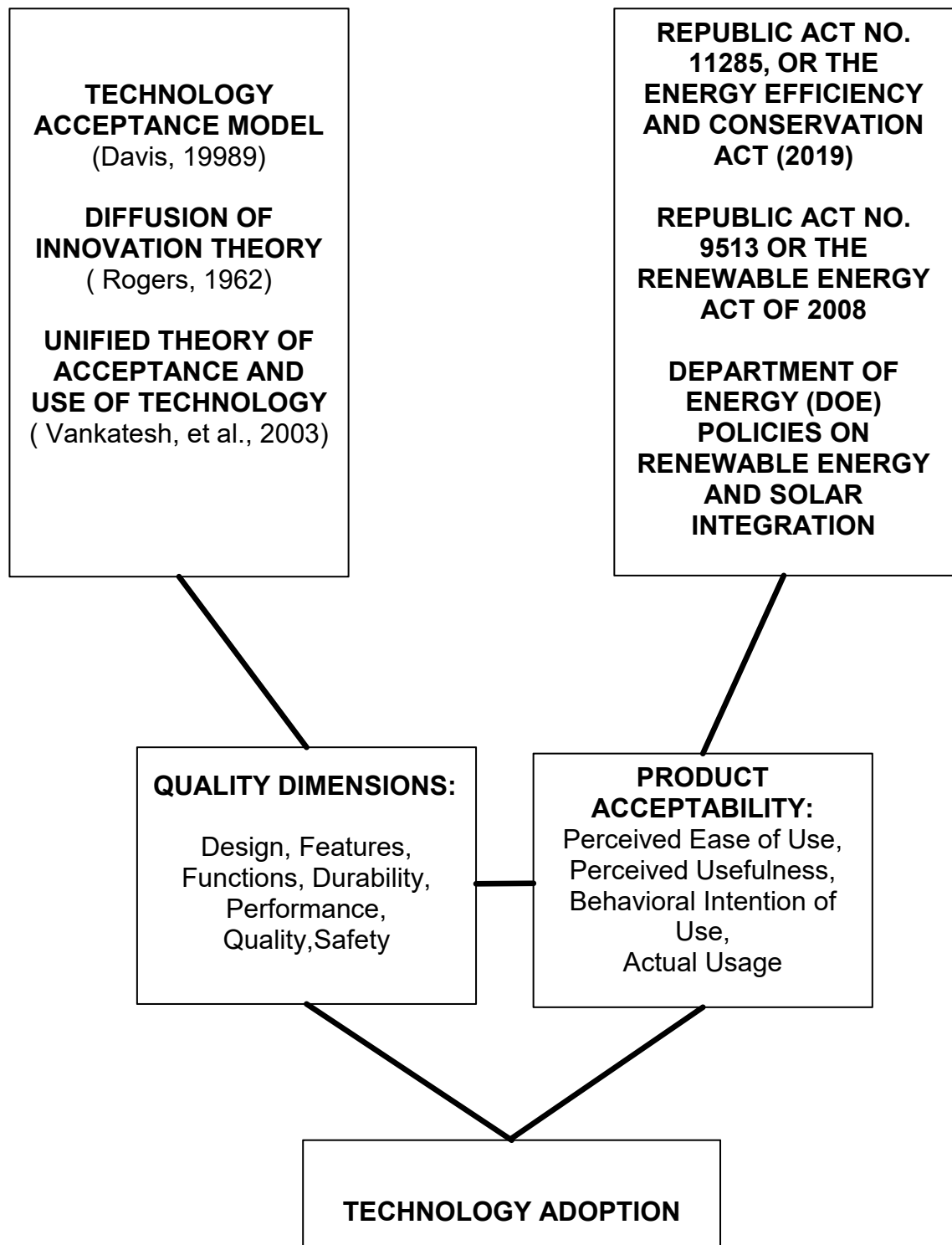
In this light, the researchers wanted to conduct this study to develop a monitoring system of a solar photovoltaic array at Cebu Technological University – Tuburan Campus. This enables real-time monitoring and management of PV system performance, allowing for prompt detection of faults, inefficiencies, or energy losses. Additionally, it aims to examine the effectiveness and acceptability of the device among the users.

Theoretical Background

The research is anchored on three theories that support and strengthen the investigation of the study. The Technology Acceptance Model (TAM) proposed by Fred Davis (1989), the Unified Theory of Acceptance and Use of Technology (UTAUT) developed by Venkatesh et al., (2003), and Diffusion of Innovation Theory by Rogers (1962).

This study is supported by the Technology Acceptance Model Theory by Davis (1989). This theory explains how users come to accept and use a new technology. This theory has four significant concepts; perceived usefulness, perceived ease of use, behavioral intention of use, and actual usage. It emphasizes two significant factors: perceived usefulness that focuses on the belief that using the technology will enhance performance and efficiency, and perceived ease of use that explains that the technology is

Figure 1. Theoretical Framework



simple and convenient to operate. The user's behavioral intention to use the technology is strongly influenced by these perceptions, which reflects their willingness to adopt it. Over time, this intention leads to the actual use of the system, where the technology becomes integrated into regular practice. The primary objective of TAM was to shed light on the processes underpinning the acceptance of technology, in order to predict the behavior of and provide a theoretical explanation for the successful implementation of technology. Therefore, TAM suggests that if a system is both useful and easy to use, people are more inclined to accept and utilize it in real-world scenarios.

The photovoltaic monitoring system is highly related to TAM: in this area irrigation can be kept stable and economical by allowing the solar water-pumping system to pump water more strenuously the better its performance. Research indicates that IoT-based solar monitoring strikingly enhances system reliability (fault detection ahead of time), first and foremost water usage below set parameters in fact near continuous cream otherwise under tremendous conditions without any external input or regular cleaning EWH. For example, Gimeno-Sales et al (2020) show in detail that PV monitoring can contribute to enhancing water pumping standards, whereas while Islamiati et al. (2022) or any other literatures do not actually indicate what the pump is doing to the fluid in terms of benefiting farmers who are actively engaged with their crops. Salman and Lam (2020) found that Internet-of-Things (IoT) systems enable solar power pumps to function more efficiently than they could in agricultural or hydroponic environments. These studies back TAM's deployment of a monitoring system to improve operational continuity, lower

energy expenses, and advance technology-driven, sustainable water management.

Another theory that anchored on the study is The Unified Theory of Acceptance and Use of Technology (UTAUT), proposed by Venkatesh et al. in 2003, which aims to provide a comprehensive understanding of technology acceptance and use. It integrates key constructs that predict behavioral intention and actual use of technology. It highlights four primary determinants of user acceptance: performance expectancy, or the belief that the technology will improve efficiency and outcomes; effort expectancy, or the perceived ease of use; social influence, which reflects the pressure or encouragement from significant others to adopt the technology; and facilitating conditions, which refer to the availability of organizational and technical support necessary for its use. UTAUT extends TAM by adding social and organizational factors, showing that people adopt technology not only because it is useful and easy to use but also because of social pressures, available resources, and support systems.

This study is significant because it addresses both the technological and human factors involved in implementing a photovoltaic (PV) array monitoring system. By using the Unified Theory of Acceptance and Use of Technology (UTAUT) as a framework, the research examines how users are likely to adopt and use the system effectively. In order to ensure the system's sustainability and useful advantages for the school, behavioral intention and actual use are shaped by performance expectancy, effort expectancy, social influence, and facilitating conditions (SpringerLink, 2023). This also shows that UTAUT effectively predicts technology adoption in renewable energy

systems (Ramli, 2025). Additionally, awareness has been found to be a significant factor in determining the intention to use solar energy, highlighting the need of user education and involvement in fostering technological acceptance (Aravindan et al., 2022).

In addition, the study draws insights from the Diffusion of Innovation Theory (Rogers, 1962), which explores the process of how new ideas and technologies are spread within a society. This theory posits that the adoption of an innovation is influenced by its relative advantage, compatibility, complexity, trialability, and observability. These certain characteristics would simply contribute to a successful adaptation of an innovation. This theory identified five key characteristics of innovations that influence their rate of adoption. The first characteristic is relative advantage, which refers to the degree to which an innovation is perceived as superior to existing alternatives. Innovations that offer clear advantages are more likely to be adopted. Compatibility is the second characteristic, focusing on the extent to which an innovation is perceived as consistent with existing values, needs, and practices of potential adopters. Innovations that align well with established norms and practices are more likely to be embraced. The third characteristic is complexity, which refers to the perceived difficulty of understanding and using the innovation. Innovations that are easier to comprehend and utilize are more likely to be adopted. Trialability is the fourth characteristic, emphasizing the ability to experiment with the innovation on a limited basis before making a full commitment. Innovations that can be tested and tried out are more likely to gain acceptance. Lastly, observability is the visibility of the innovation's benefits and results to others. Innovations that are easily

observable and have tangible benefits are more likely to be adopted, as they provide social proof and encourage further adoption.

The Diffusion of Innovation Theory is highly relevant to the development of Photovoltaic (PV) array monitoring systems to characteristics such as relative advantage, compatibility, complexity, trialability, and observability (Oliva & Atehortua Santamaria, 2025). These elements affect early adopters and the speed at which users incorporate the technology. Using DOI aids in the development of adoption-boosting tactics like training, pilot testing, and transparent performance reporting (Jones & Warren, 2020). All things considered, DOI offers a framework for encouraging the adoption and efficient use of PV monitoring systems (Oliva & Atehortua Santamaria, 2025).

This study is also supported by a legal basis through Republic Act No. 9513 or the Renewable Energy Act of 2008, which promotes the development and utilization of renewable energy resources in the Philippines. By implementing an IoT-based solar monitoring system, the device contributes directly to the goals of RA 9513: reducing dependence on fossil fuels, mitigating climate change, and promoting sustainable energy practices in higher education. This legal foundation strengthens the relevance of the study, linking it to national policy and long-term sustainability goals.

This highlight the relevance of this legal basis by offering financial incentives like tax holidays, duty-free equipment imports, and net-metering, Republic Act No. 9513 (Renewable Energy Act of 2008) encourages the use of solar PV systems (LawPhil, 2008). By lowering financial obstacles, these incentives increase the viability of solar array and monitoring system

purchases. According to research, net-metering regulations increase rooftop solar PV projects' economic viability (Chiu et al., 2020). Additionally, policy incentives promote the wider use of solar PV systems in developing nations (Saleh et al., 2022). Additionally, regulatory frameworks are crucial in advancing smart solar energy technologies that improve system performance and reliability, like real-time monitoring systems (Frenzel et al., 2023).

This study is supported by the Republic Act No. 11285, or the Energy Efficiency and Conservation Act (2019), aims to promote efficient use of energy across all sectors in the Philippines. It establishes policies, programs, and incentives to reduce energy consumption, improve energy efficiency in buildings, industries, and appliances, and encourage the adoption of energy-saving technologies. The law supports sustainable energy practices and helps lower greenhouse gas emissions while reducing costs for consumers and businesses.

Republic Act No. 11285 encourages the use of energy-saving technologies and energy efficiency. By lowering losses, increasing overall efficiency, and enabling real-time surveillance and management of solar energy production, photovoltaic (PV) array monitoring systems help achieve this objective. According to Yazdani et al. (2023), smart monitoring systems can improve PV performance by identifying inefficiencies like hot spots. According to Medina-García et al. (2021), wireless monitoring and control systems maximize energy output while preserving system stability. Ansari et al. (2021) emphasize how communication protocols and data-processing technologies can improve PV system performance. Lastly, Artono et al. (2023)

demonstrate how real-time optimization and improved energy management are made possible by IoT-based monitoring.

This has been also supported by Department of Energy (DOE) Policies on Renewable Energy and Solar Integration. The DOE policies on renewable energy and solar integration provide guidelines and regulations to ensure the safe, efficient, and standardized implementation of renewable energy systems in the Philippines. These regulations seek to support national objectives for the development and dependability of renewable energy sources, maximize energy output, and encourage sustainable energy practices.

This highlighted the relevance of this policy to the study. A framework for the safe, effective, and uniform operation of solar energy systems is provided by the Department of Energy's (DOE) policy on renewable energy and solar integration (Department of Energy [DOE], 2015–2023). According to Yazdani et al. (2023), smart monitoring systems can improve energy efficiency by identifying inefficiencies like hot spots. Wireless monitoring and control systems maximize energy output and preserve system stability, according to Medina, et al. (2021). Ansari, et al. (2021) highlight how different monitoring methods and protocols can improve the performance of PV systems.

In conclusion, the incorporation of theories and legal basis support and strengthen the foundation for the study. The integration of these ensure that IoT-monitoring systems for solar photovoltaic arrays comply with safety, data privacy, and energy regulations, providing a secure and standardized approach to renewable energy management..

Statement of Problem

This study aimed to assess the efficiency and acceptability of the Solar Photovoltaic Array Monitoring System for cost efficiency and effectiveness regarding water supply at Cebu Technology University during the academic year 2025-2026, as a basis for technology adoption.

Specifically, it sought to answer the following questions:

1. What are the existing prior arts of a Photovoltaic Array Monitoring system?
2. What are the technical requirements in developing an of Solar Photovoltaic Array Monitoring System in terms of;
 - 2.1 detailed device plan;
 - 2.2 material cost; and
 - 2.3 safety and maintenance?
3. What is the level of effectiveness of the of Solar Photovoltaic Array Monitoring System in terms of:
 - 3.1 performance;
 - 3.2 function;
 - 3.3 convenience;
 - 3.4 serviceability; and
 - 3.5 perceived quality?
4. What is the level of user acceptance and satisfaction of the of Solar Photovoltaic Array Monitoring System based on:
 - 4.1 perceived usefulness;
 - 4.2 perceived ease of use;
 - 4.3 behavioral intention to use; and
 - 4.4 actual usage?

5. Is there a significant relationship between the perception of the respondents on the level of effectiveness and acceptability of the developed product?

5. Based on the findings, what technology adoption could be proposed?

Null Hypothesis

H0. There is no significant relationship between the perception of the respondents on the level of effectiveness and acceptability of the developed product.

Significance of the Study

The following identifies the main beneficiaries of the study and how each benefits from it.

Students in CTU-Tuburan. This study may benefit the students by providing access to accurate, real-time data on their solar power usage and performance, enabling them to manage electricity consumption efficiently and reduce electricity costs.

Educational institutions. This study may benefit various schools in employing this research as a reference and instructional tool for students studying renewable energy and Internet of Things (IoT) applications.

Solar power system users. These people might gain from the monitoring system in that it helps them identify problems, improve performance, and save money on maintenance.

Technology Students. This study may improve their technical abilities for upcoming advancements in sustainable technologies by acquiring practical knowledge and understanding of how the Internet of Things (IoT) is integrated into renewable energy systems.

Local Government Unit of Tuburan. This study may become beneficial developing policies that support environmental sustainability, encouraging the use of renewable energy, and putting smart energy management systems into place.

CTU - College of Technology. This may benefit the department in integrating prevalent innovations towards sustainable renewable energy. This gives them the idea of how to strengthen research and innovation for the students and community as a whole.

Future Researchers. Future innovators may benefit by having a foundation for further advancements in developing a sustained renewable energy using photovoltaic arrays with IoT monitoring this study encourages sustainable energy habits, supports environmental preservation, and uses intelligent technology integration to advance community development.

Scope and Limitations of the Study

This study focuses on designing and developing an IoT-based monitoring system for solar photovoltaic (PV) arrays at Cebu Technological University (CTU) - Tuburan Campus. The study will be conducted during the fiscal year 2025-2026 and will end by the close of the year. The study will focus on gathering data from a sample of respondents from the CTU-Tuburan community such as students, faculty, and staff, will take part in data gathering. to assess their perceptions of the IoT-based monitoring system for solar photovoltaic (PV) arrays usability, effectiveness, and overall acceptance. By gathering this information, researcher will be able to detect potential issues and improve the product's acceptability and functionality for future users. The

research instrument that will be utilized in the study is a modified questionnaire which will undergo validity and reliability.

Furthermore, the study is equipped with key features that monitors and displays key information from the solar photovoltaic array. It shows important data such as voltage, current (ampere), temperature, energy consumption per hour and per day, and water level consumption per day. Users can view this information remotely through an easy-to-use dashboard in an application and in person. The system provides real-time monitoring and data display. Using affordable components like ESP32 microcontrollers and low-power sensors makes the system practical and suitable for use. Though, it uses affordable parts like low-power sensors, it may face issues like Wi-Fi signal problems or sensor wear during heavy rains.

The research data gathering relies on surveys and basic tests, but it does not use advanced tools like full simulations or weather impact studies due to time and budget limits. This study also limits to small campus or small residential set ups and maybe enhanced to a broader scale. On the other hand, this does not send and notify message on the user side specifically the features display.

Definition of Terms

The following terms are defined to present a higher level of comprehension and guidance as they were utilized in the study.

Actual Usage. This refers to the real-world use of the IoT monitoring system for checking solar PV arrays, based on how often and how well they interact and perceive its features .

Convenience. As used in the study, this refers to how simple and practical the IoT monitoring system is for the users to access and operate, such as through mobile apps or quick data views, making solar checks faster and less effort.

Design. This refers to the materials being used in creating this IoT monitoring system and how it would look like at the end.

Efficiency. This refers to how well the solar PV arrays produce and use energy after the IoT system spots problems early.

ESP32 Microcontroller. This refers to the affordable, low-power chip used in the IoT system to connect sensors, process data, and send updates, making the setup practical and easy for solar monitoring.

Function. This refers to the working parts and tasks of the IoT monitoring system, such as measuring voltage and current, to ensure solar PV arrays run smoothly in classrooms and labs.

Internet of Things (IoT). This refers to the network of connected devices, like sensors and apps, that gather and share solar data in real time, and improve monitoring without big costs.

Maintenance Program. This refers to the planned steps to keep the IoT system and solar panels running well long-term, including checks, fixes, and updates for safety and efficiency.

Monitoring System. This refers to the system that monitors and keeps track of the data recorded and flashed on the dashboard with IoT setup.

Performance. This refers to how does the device function. It is also carrying out the over-all capacity of the device on how it is being accepted by the user.

Photovoltaic (PV) Arrays. It refers to the groups of solar panels used to turn sunlight into electricity for buildings, which the IoT system monitors to keep them efficient.

Real-time Monitoring. As used in the study, this refers to the live tracking of solar data, like voltage and faults, sent instantly to users via the IoT system, allowing quick fixes to avoid energy loss.

Serviceability. This refers to how easy it is to repair and maintain the IoT system, such as swapping sensors or updating software, to keep it working in the campus environment.

Sensors. This refers to the small devices in the IoT system that measure things like current, temperature, and sunlight intensity on solar panels, providing key data.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the relevant literature and studies that the researcher considered in strengthening the importance of the study. It also presents the synthesis of the art to fully understand the research for better comprehension of the study.

Prior Arts

Solar energy has become an increasingly popular source of renewable energy due to its sustainability and environmental benefits. Over the years, researchers and engineers have focused on maximizing the efficiency and reliability of solar photovoltaic (PV) systems by incorporating advanced technologies. One key development is the integration of the Internet of Things (IoT) in monitoring solar PV arrays. This integration allows for real-time data collection, analysis, and remote management, which significantly enhances the system's performance and maintenance. Moreover, the use of IoT in solar PV monitoring supports sustainability efforts in educational institutions and remote locations. There are many communities benefit from such technologies by ensuring their renewable energy installations operate efficiently while educating students on cutting-edge technology applications.

Narvios & Nguyen (2021) studied the Internet of Things (IoT) based system was made to monitor, detect dust accumulation, and a cleaning system that would automatically wipe the dust on the surface of the PV solar panels. Using a specific dust sensor, it detects and monitor the amount of dust in the panel. As the dust accumulated reaches its limit, the cleaning system triggers automatically. The ambient temperature was monitored using

the temperature and humidity sensor. In addition, a device created by Tsamaase et al., (2017) also emphasized on development of an automated dust detection and cleaning system which could be used to clean photovoltaic (PV) modules. It showed that the system was able to identify power loss brought on by dust buildup on the module surface, and the cleaning mechanism's motor drive responded as a result in order to run the cleaning mechanism.

In the study conducted by Mantua et al.,(2024),it emphasized that the goal of this device is to provide smart water irrigation for rice farming by utilizing microcontroller and Internet of Things devices that can support solar panels. Through intelligent temperature, soil moisture, and humidity monitoring utilizing several sensors, the system reduces physical work and exemplifies the potential of automated irrigation. This resulted to a very practical and valuable innovation. Moreover, Solar-Powered IoT Water Quality Monitoring System also employed solar energy to power sensors that measure temperature, turbidity, and pH. For continuous and sustainable water quality testing, particularly in remote places, the system sends data to an online portal for real-time monitoring (Sheikameer et al.,2024).

In a study conducted by Patil and Patil (2020) produced an Internet of Things-based solar power monitoring system that uses a cloud-based interface to gather and send photovoltaic performance data including voltage, current, and power production. Their research emphasizes how IoT improves solar system energy management and early failure identification. Furthermore, to promote sustainable environmental management, Gómez et al.,(2021) emphasized an Internet of Things-enabled environmental monitoring system

that tracks temperature, humidity, and air quality and is powered by renewable energy sources. When taken as a whole, these studies show how IoT combined with renewable energy technology can facilitate effective, real-time monitoring systems that can be used for energy-based and environmental applications.

Certain areas suffer frequent energy shortages due to their limited access to the power grid. In this case, Narvios et al., (2021) the study says that a certain areas suffer frequent energy shortages due to their limited access to the power grid. With this notion, it is very recommended that these areas utilize alternative forms of producing electric energy. The Solar Home System (SHS) has helped millions of families gain access to electricity in Asia and requires the simplest form of a panel and battery. Using sensors, controllers, Wi-Fi modules, and software applications through Blynk, an Internet of Things (IoT)-based swarm solar household system was successfully obtained for this study. It has an integrated cloud server for monitoring and controlling the connection between families, and energy sharing is enabled. Similarly, Renewable energy based rural electrification system experiences fluctuation in generation efficiency due to various unpredictable dynamic changes such as solar irradiation levels, rainfall, dust accumulation, vegetation, etc. This study proposed a micro grid cluster connecting technology to increase system efficiency. The researchers have created an Internet of Things system that can monitor and learn dynamic changes in power quality to regulate the subsystems for enhancing the quality of service in order to develop real-time judgments on energy management and micro grid interconnections (Shibu et al., 2018).

On a study by Rani et al., (),they developed a device Automated alternative energy consumption from any location via the internet is made possible by a machine-controlled Internet of Things (IoT), which is essentially an alternative energy viewing device. In order to meet the requirements for solar batteries, select AT super controllers, with a primary concentration on mega controllers. Our technology sends the capacity performance of the IoT system across the internet and continuously monitors the solar battery. We routinely transmit alternative energy parameters to the IoT Thing Speak server's network via IOT Thing Speak. It now warns the user when the performance limit is reached and shows these metrics in an easy-to-use graphical user interface. This guarantees optimal power generation and makes it simple to observe star plants remotely.

Similarly, as innovated by Datar et al., (2023),solar plants were to build in the locations where people can't reach on daily basis so this approach will help them to virtually control their systems from faraway places. It monitors the panel load by using the IoT technologies and the data which are received from the panels are send to the cloud through the internet for the future use. It also helps the remote users to monitor the solar power plant. This system is designed to solve the problem occur in solar power generation like management problem, maintenance and to reduce the time of repair. Using this technology, the cost of solar energy (renewable energy) generation reduces. This also provide real time information to the user help to monitor the system.

Ramu et al., (2021). presented an Internet of Things (IoT) technologies with smart sensors play a vital role in monitoring and control applications in

many areas. This explores how to monitor the solar Photovoltaic (PV) system using IoT, and addresses various remote monitoring methods. It elaborates on the real-time implementation of smart sensors used in the renewable energy system and its challenges with some solutions. A solar cell or photovoltaic cell is designed to observe solar energy and produce electric power.

Similarly, Tellawar & Chamat (2019) developed an IOT based smart solar photovoltaic remote monitoring system which is a low-cost, user-friendly, reliable data logger and monitoring system has been developed mainly for a pico solar home system in a rural area of a developing country. This ESP 32 microcontroller based data logger stores all monitoring parameter in a micro SD card and displays that on a Blynk App. Data can be downloaded directly from the webpage to analyze and verify the system operation. The developed data logger hardware prototype uses only four sensors for humidity, temperature, voltage, and current sensing. An already developed Android app is use for cell phone to display all parameters in real time basis for an efficient monitoring which can also able to give important information to maintenance personnel for any issues in battery charging. To accomplish the task, we have developed the hardware implementation of remote data acquisition architecture of photovoltaic systems based on the Internet of Things. The overall cost of this prototype is economical.

Recent studies and industry reports highlight the increasing use of IoT technology for photovoltaic (PV) system monitoring, showcasing both innovative research and useful applications. IoT-based monitoring can improve PV system performance by facilitating real-time data collection, fault

detection, and predictive maintenance, according to academic studies.

In the report conducted by Palak (2025), the solar energy sector is undergoing a technological transformation, where photovoltaic (PV) systems are no longer passive energy generators but intelligent assets capable of real-time monitoring, diagnostics, and optimization. The integration of Internet of Things (IoT) technologies with solar panels is reshaping how residential, commercial, and utility-scale systems are managed, making operations more efficient and financially predictable. Thus, IoT-enabled smart solar panels are revolutionizing the solar energy landscape by providing real-time performance tracking, predictive maintenance, and operational efficiency across residential, commercial, and utility segments. As sensor costs decline, connectivity improves, and analytics platforms become more sophisticated, smart monitoring is set to become the industry standard, maximizing returns for investors and accelerating the adoption of clean, reliable solar energy.

Similarly, Rayzon Solar (2025), emphasized that solar energy is known for its clean and renewable nature, has become a cornerstone in combating climate change. When paired with IoT, solar systems become even more powerful. IoT integrates sensors, devices, and communication technologies to enable real-time monitoring, data analysis, and automation. This fusion allows for smarter energy management, ensuring maximum efficiency and minimal waste. It's clear that the integration of IoT with solar energy systems will play a pivotal role in achieving global energy goals. Embrace the power of these technologies today and be part of a cleaner, smarter energy future.

Robustel (2022), explained how Internet of Things (IoT) technologies have a big impact on solar panel system management and monitoring. It

highlights how IoT makes it possible to monitor individual panels in real time. This covers external elements like dirt or debris on the panels and external damage, as well as internal factors like temperature, irradiance, voltage, and current. Decisions about operations and management are based on this data. Regardless of their resources or budget, energy businesses may find it difficult to collect it. The application of IoT also explores in both industrial and residential settings. IoT in industrial applications enables thorough monitoring of massive solar installations, giving operators in-depth knowledge of system performance. From smart software to connected devices, IoT solar panel monitoring is helping businesses and residents monitor how much energy is being generated and how much is being consumed. Consumers can also discharge any excess electricity into the smart grids, in exchange for money. IoT-enabled solar panel monitoring systems can improve the overall sustainability and efficiency of solar energy systems by alerting users of any problems before they become more serious.

Another significant discovery from CXOToday(2025) is Smart Solar Adoption: Leveraging IoT and Real-Time Monitoring for Better Energy Outcomes. Smart solar adoption, empowered by the Internet of Things (IoT) and real-time monitoring is one of the renewable energy solutions. Smart solar systems are not just solar panels; they're part of a connected ecosystem. These systems integrate IoT-enabled sensors, meters, and controllers directly into PV installations. These devices continuously collect and transmit data on vital parameters such as: panel temperature and efficiency, shading impact and dust accumulation, inverter performance and battery charge/discharge cycles, real-time power generation and consumption. More than just a

technical advancement, the combination of solar energy, IoT, and real-time analytics is a paradigm shift in the way we sustainably manage energy. Both private users and utility-scale suppliers are empowered by smart solar adoption, which offers improved performance tracking, predictive maintenance, and seamless grid integration. Smart solar systems will be crucial to achieving sustainability goals as well as creating a resilient, responsive, and intelligent energy future as the world's energy needs increase and the urgency of the climate crisis increases.

Njoka et al., (2023) emphasized that the deployment of remote monitoring systems based on Internet of Things (IoT) presents an opportunity to curtail operational and maintenance costs associated with stand-alone PV systems. This study evaluates the characteristics of the commonly employed IoT platforms, their capabilities and associated O&M cost savings. Analysis of avoided field visit costs based on three remotely monitored solar PV sites is conducted through clustering of system faults and filtering out major ones that would warrant actual site visits.

Portlandia Energy Supply (2025) has managed and optimized energy production has been completely transformed in recent years by the integration of IoT technology with solar energy systems. We can improve system performance and energy efficiency by using IoT devices to monitor and regulate solar panels in real-time. In addition to optimizing energy output, this creative strategy aids in the shift to more ecologically friendly and sustainable energy sources. In commercial and industrial settings, their Smart Energy Management Systems (SEMS) allow for automatic energy control and real-time monitoring. In order to reduce wasteful use and enhance sustainability,

these systems make use of cloud-based platforms and IoT-enabled software to offer comprehensive insights into energy consumption, spot inefficiencies, and modify energy flows in response to demand. Additionally, they provide AI-powered solar inverters that balance production and consumption, forecast daily energy output, track panel performance, and facilitate predictive maintenance to cut expenses and downtime.

As reported by Kahana (2023), Researchers from Egypt have utilized NodeMCU WiFi modules and a single-board Raspberry Pi to monitor and control PV modules. Their tests showed a “good degree of agreement” between the system and multimeter readings. It combines a custom-made printed circuit board (PCB) attached to the PV module and a single-board Raspberry Pi computer that communicates wirelessly over WiFi. According to the research group, the PCB of the system includes three NodeMCU WiFi modules that allow it to communicate with Raspberry Pi. One of those modules is in charge of collecting measurements, the second for servo motor control, and the third for On/Off control. The latter control can be used for various applications, such as battery storage, cleaning systems, and cooling pumps.

Zeversolar (2016), a subsidiary company of leading inverter manufacturer SMA, has launched a new app for monitoring PV systems via smartphone. The app is suitable for private and commercial users and, like the monitoring system via ZeverCloud, free of charge. It shows daily, monthly, and annual yields clearly in compact diagrams and also provides information about feed-in tariffs generated. New systems are registered quickly and easily when users scan a QR code on their Zeversolar inverter with a smartphone.

The inverter is subsequently connected to the internet router via WLAN or a LAN cable. The most important part of monitoring is giving operators reliable information about possible yield losses. Therefore, our monitoring software uses the cloud. If a server fails, another automatically takes over its tasks. With ZeverCloud, yield losses can be identified partly by the comparison of current and previous yield data. In addition, the software sends error messages by e-mail.

It was clear that the aforementioned related literature and studies support and strengthen the idea of having a sustainable renewable energy. Several studies have shown the increasing relevance of IoT in solar energy systems, and photovoltaic power. Thus, there is insufficient evaluation of system reliability and accuracy under extreme weather conditions, partial shading, and tropical climates. Therefore, developing a simple, cost-effective IoT monitoring system for PV arrays is important particularly in real-world conditions. It truly enhances energy efficiency, reduce operational costs, and support the sustainable expansion of solar energy.

CHAPTER 3

METHODS AND MATERIALS

This chapter presents the methodology used in this study. It discusses the steps undertaken to design, develop, and evaluate an IoT monitoring system for the solar photovoltaic array at Cebu Technological University – Tuburan Campus. This chapter includes the research design, study flow, environment, respondents, instruments, data gathering procedure, and data analysis techniques.

Research Method

This study employed developmental research and descriptive-correlational types of research. According to Siedlecki (2020), the descriptive approach aims to accurately and systematically describe a population, situation or phenomenon. This approach is applied in presenting and discussing the development of an IoT-based monitoring system for solar photovoltaic (PV) arrays among the respondents description in terms of the effectiveness (design, performance, functions, convenience, serviceability, perceived quality), and the products acceptability in terms of perceived usefulness, perceived ease of use, behavioral intention of use, and actual usage. Correlational research is a methodological approach that aims to identify and analyze the relationship between two or more variables without manipulation (Putri et al.,2025).

According to Ibrahim (2016), developmental research is oriented toward a general analysis of design, development and evaluation process addressed either as a whole or as a particular component. Thus, this research method will be employed when creating and developing the product which is

an IoT-based monitoring system for solar photovoltaic (PV) arrays. To obtain the data from the respondents, the researchers will use a standard procedure to ensure each respondent can answer the questions at a level of fairness, to avoid biased opinions that could influence the outcome of the study. The process involves asking people for information through a survey questionnaire where they also get to see and use the product in real setting.

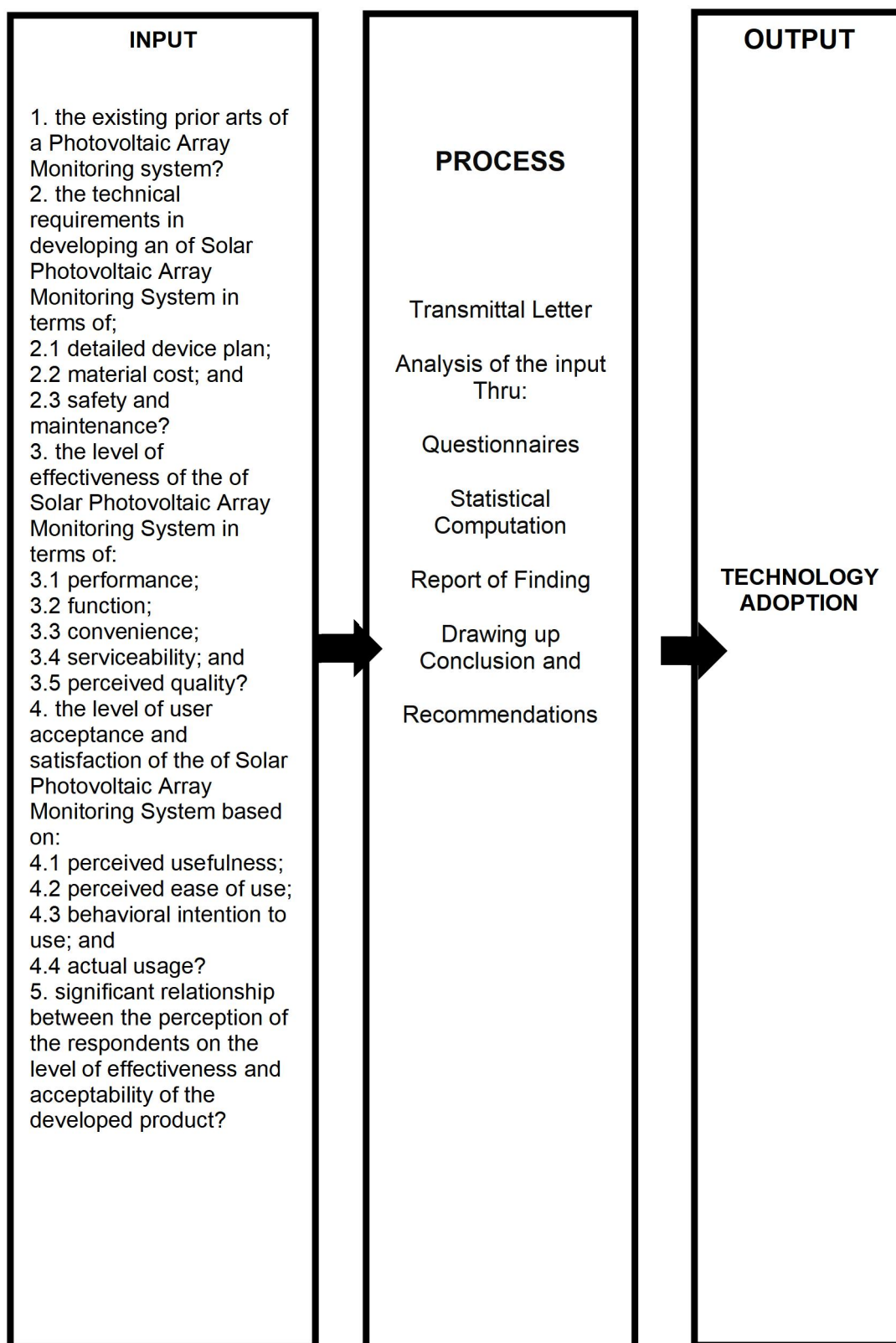
Flow of the Study

As what have shown in Figure 2, it clearly presents how the study flows. As shown, a composition of input, process, and output. The input includes the profile of prior arts for existing solar monitoring technologies, technical requirements for the IoT system, the effectiveness of the product in terms of (function, convenience, serviceability, performance, and quality), and the products acceptability in terms of perceived usefulness, perceived ease of use, behavioral intention of use, and actual usage.

The evaluation will be conducted through the distribution of questionnaires to the respondents with the signed approval letters from the respective authorities. This part belongs to a certain process where the collection is going to happen. After accumulating the data, the researchers will interpret it by using tables and proceed to the findings, drawing up conclusions and recommendations.

Moreover, the output refers to the result of the study, which is the technology adoption. The IoT system makes solar monitoring easier and safer due to added features like sensors for voltage, current and temperature, flow-meter, real-time monitor, weatherproof casings, and user-friendly dashboards with ESP32 microcontrollers. This also has a system that monitors the water

Figure 2. Flow of the Study



level of the deep wheel. This emphasizes that this device needs to have a concrete and developed maintenance program to highlight the importance of an IoT monitoring system in order to meet the energy needs in a specific setting for technology adoption.

Research Environment

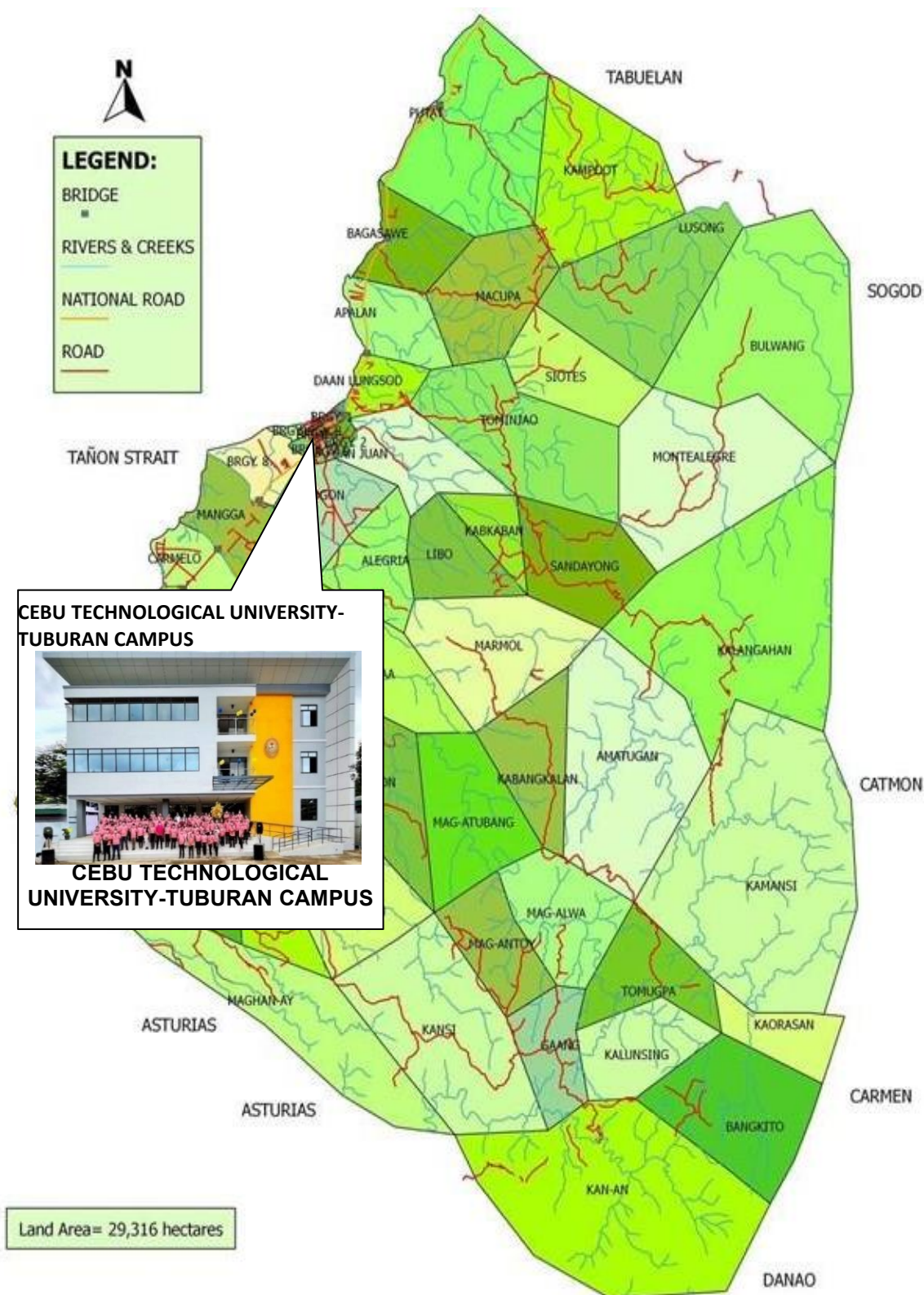
Cebu Technological University (CTU) Tuburan started its humble beginnings in 1945. It is situated in Brgy. 8, Tuburan, Cebu. The campus has a land area of 79, 936 sq meters. In the academic year 2019-2020, there are 3066 undergraduate students and 130 graduate students respectively.

There are six colleges in CTU Tuburan, with each offering different degree programs. The College of Engineering offers Bachelor of Science in Civil Engineering, Bachelor of Science in Electrical Engineering, Bachelor of Science in Industrial Engineering and Bachelor of Science in Mechanical Engineering. The College of Arts and Sciences offers two-degree programs : Bachelor of Arts in English Language (BAEL) major in English Language Studies and Bachelor of Arts in Literature (BaLit) major in Literary and Cultural Studies. The College of Technology (COT) offers Bachelor of Industrial Technology with nine majors: Automotive Technology, Computer Technology, Drafting Technology, Electrical Technology, Electronics Technology, Food Preparation Services Technology, Garments Technology, Machine Shop Technology and Welding and Fabrication Technology. Additionally, COT also has Bachelor of Science in Information Technology and Bachelor of Science in Hospitality Management. The College of Education offers three-degree programs: Bachelor in Elementary Education, Bachelor in Secondary Education major in Mathematics and Bachelor of Technology and Livelihood

Education. The College of Agriculture offers Bachelor of Science in Agriculture with three majors: Animal Science, Agronomy and Horticulture. The Graduate School offers Doctor in Development Education, Master in Public Administration, Master of Arts in Vocational Education and Master of Arts in Education major in Administration and Supervision (CTU-Tuburan,2025).

The researchers chose Cebu Technological University Tuburan Campus as the locale of the study because the school has been a great avenue for innovation. Furthermore, to ensure consistency in innovation, this device offers quality and efficiency in order to widen knowledge and to develop the skills of the students. Moreover, this would be beneficial in the institution itself since this could help boost energy efficiency and monitors the water supply.

Figure 3. Locale of the study



Research Respondents

The respondents of this study were composed of students, and maintenance staff from Cebu Technological University (CTU) Tuburan Campus. These respondents were chosen because they have knowledge or experience related to the field. In addition, the students were included because of their academic and technical background, maintenance staff were considered for their hands-on role in the system. The researchers used purposive sampling in selecting the respondents.

Table 1
Distribution of Respondents
n=30

Respondents	f	%
Students	15	50
Maintenance Staff	15	50
Total	30	100

Table 1 shows the distribution of respondents for the study. As presented in the table, fifteen or 50 percent of the respondents were students, fifteen or 50 percent of the respondents were maintenance staff. This gives a total of fifty (30) respondents or 100 percent from CTU Tuburan Campus.

Research Instrument

The study utilized a modified questionnaire from Sumbi et al.(2024) entitled “Community Based Multipurpose Charging Station: Technology Adaptation” specifically in Part 1 of the questionnaire. Part I collected information on respondents' feedback on the level of effectiveness of the product in terms of function, convenience, serviceability, performance, and

quality. The product's acceptability is tailored from the Technology Acceptance Model of Davis (1986), specifically in Part 2. Part 2 focused on the product's acceptability in terms of perceived usefulness, perceived ease of use, behavioral intention of use, and actual usage. The questionnaire underwent pilot testing, and three experts evaluated the validity and reliability of the questionnaire.

This study used a survey questionnaire as the main tool in gathering data from the selected respondents of Cebu Technological University (CTU) Tuburan Campus. The respondents included students, and maintenance staff since they are directly involved in the use, monitoring, and management of solar panel systems on the campus.

Research Procedure

This procedure outlines the systematic steps involved in constructing and implementing an IoT-Based Solar Photovoltaic (PV) Monitoring System, emphasizing the accurate measurement of solar parameters, efficient data transmission, and real-time monitoring.

Step 1. Materials

Prepare all required components, tools, and safety accessories to ensure smooth assembly and operation of the IoT-Based Solar Photovoltaic (PV) Monitoring System. All materials below were inspected prior to installation to verify functionality and compatibility with the solar power setup at Cebu Technological University – Tuburan Campus.

Table 2. Electrical & Electronic Components

Components	Quantity
ESP32 Development Board(30 pins)	1
INA226 Voltage and Current Sensor	1
DS18B20 Temperature Sensor	1
XL7015 DC-DC Buck Converter	1
YF-S201 Flow Meter Sensor	1
OLED Display (1.3" I ² C)	1
Resistors (4.7k Ω , 10k Ω)	4
Jumper Wires	1 set
PCB or Breadboard	1
Cable Glands and Stainless Bolts	Several
Solar Panel (12V or 24V)	1
Weatherproof Enclosure Box	1 box
Terminal Blocks / DC Connectors	1 set

Table 3. Tools, Safety & Installation Accessories

Materials/Tools	Quantity
Soldering Iron with Soldering Lead	1 set
Wire Stripper/Cutter	1
Multimeter/Clam Meter	1
Screwdriver Set	1 set
Drill w/ Hole Saw Bits	1 set
Electrical / Heat-Shrink Tape	1

Epoxy or Silicone Sealant	1
USB Cable (ESP32 compatible)	1
Protective Gloves & Safety Goggles	1 set
PVC Fittings for Flow Sensor	As required
Teflon Tape / Hose Clamps	Several

Step 2. Assembly of Electronic and Electrical Parts

Gather all the components needed for the system. Inspect each part to ensure functionality before assembly. Carefully connect each component according to the schematic diagram provided and secure all modules properly for safe installation.

Step-by-step wiring guide:

1. Connect the solar panel positive output to the input terminal of the XL7015 module (VIN+), and the solar panel negative output to XL7015 (GND).
2. Connect the output of XL7015 (5V OUT and GND) to the VIN and GND pins of the ESP32 to supply regulated power.
3. Connect the INA226 voltage and current sensor:
 - VCC → 3.3V (ESP32)
 - GND → GND (ESP32)
 - SDA → GPIO21 (ESP32)
 - SCL → GPIO22 (ESP32)
 - VIN+ → Solar panel positive wire
 - VIN- → Solar panel negative wire
4. Connect the DS18B20 Temperature Sensor:
 - VCC → 3.3V (ESP32)

DATA → GPIO4 (ESP32)

GND → GND (ESP32)

Add a 4.7kΩ resistor between DATA and VCC

5. Connect the OLED Display (I²C communication):

VCC → 3.3V (ESP32)

GND → GND (ESP32)

SDA → GPIO21 (ESP32, shared with INA226)

SCL → GPIO22 (ESP32, shared with INA226)

6. Install the YF-S201 Flow Meter Sensor:

Signal Wire → GPIO27 (ESP32 interrupt pin)

Red Wire → 5V (ESP32)

Black Wire → GND (ESP32)

Mount inline with water pipe using PVC fittings, ensuring alignment with water flow direction

7. Place all components securely on the PCB or breadboard, and install them inside the weatherproof enclosure using screws and supports.

8. Route all cables through cable glands and apply proper sealing to prevent water entry and ensure long-term durability.

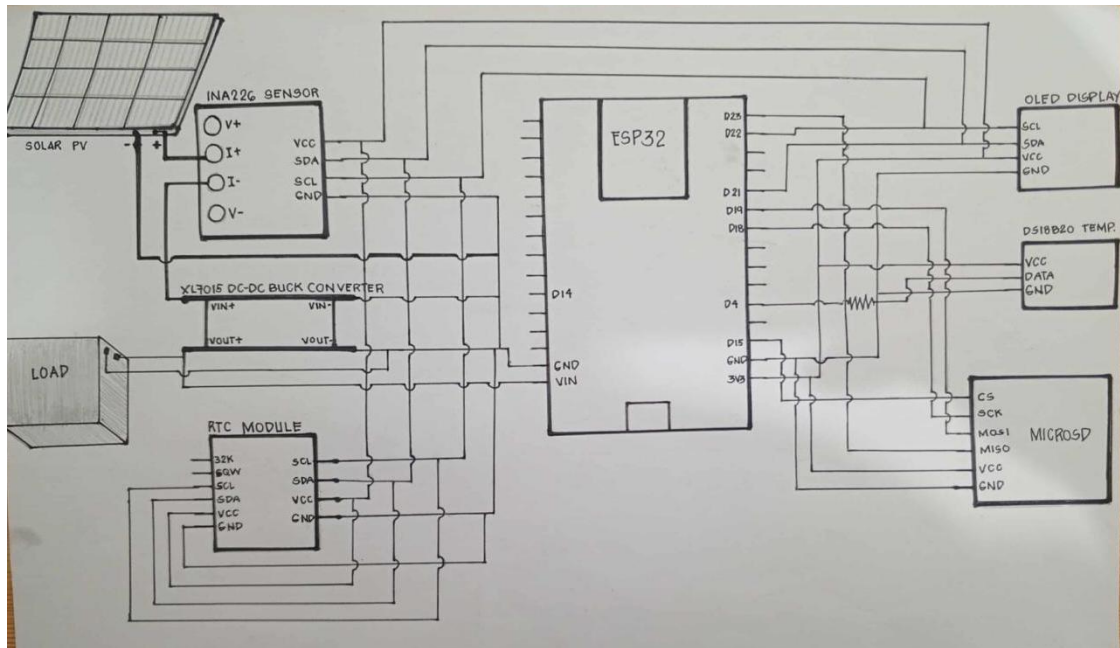


Figure 4. Product Assembling Electronic and Electrical Wiring

Step 3. Coding or Program

After connecting and assembling all components, the program for the ESP32 microcontroller is uploaded using the Arduino IDE. This program enables the monitoring system to read, process, display, and transmit data to the cloud for real-time access.

Step-by-step coding and setup guide:

1. Install and open the Arduino IDE on a computer or laptop and install the ESP32 board package through the Boards Manager.

2. Install the required Arduino libraries:

- Adafruit_INA226, Adafruit_SSD1306, Adafruit_GFX, OneWire, DallasTemperature, RTCLib, FirebaseESP32

3. Connect the ESP32 to the computer using a USB cable.

Select the correct:

Board: ESP32 Dev Module

Port: COM Port assigned by the system

4. Encode or upload the program that:

Reads voltage and current using the INA226 sensor

Reads temperature values from the DS18B20 sensor

Counts pulses from the YF-S201 flow meter to measure water flow

(Formula used: Flow Rate (L/min) = Pulse Count / 450)

5. Include power and energy computation in the program:

$P = V \times I$ (Power in Watts)

$E(\text{kWh}) = \frac{P \times t}{1000}$ (Energy consumption over time)

6. Write the display functions for the OLED module to show real-time values such as voltage, current, temperature, water flow rate, and system status.

7. Input the Wi-Fi SSID and Password into the program to enable wireless communication.

8. Connect the system to the Firebase Realtime Database and ensure the ESP32 automatically uploads time-stamped sensor data for cloud monitoring.

9. After uploading the program, verify system functionality using the Serial Monitor and OLED display to ensure all readings appear correctly.

10. Once confirmed working, disconnect the USB cable and prepare the system for field deployment.

The programmed device will now continuously gather measurements, display them locally on the OLED, and transmit them online for remote viewing of voltage, current, temperature, power generation, and water consumption of the solar water system.

Figure 5. Product Coding

```

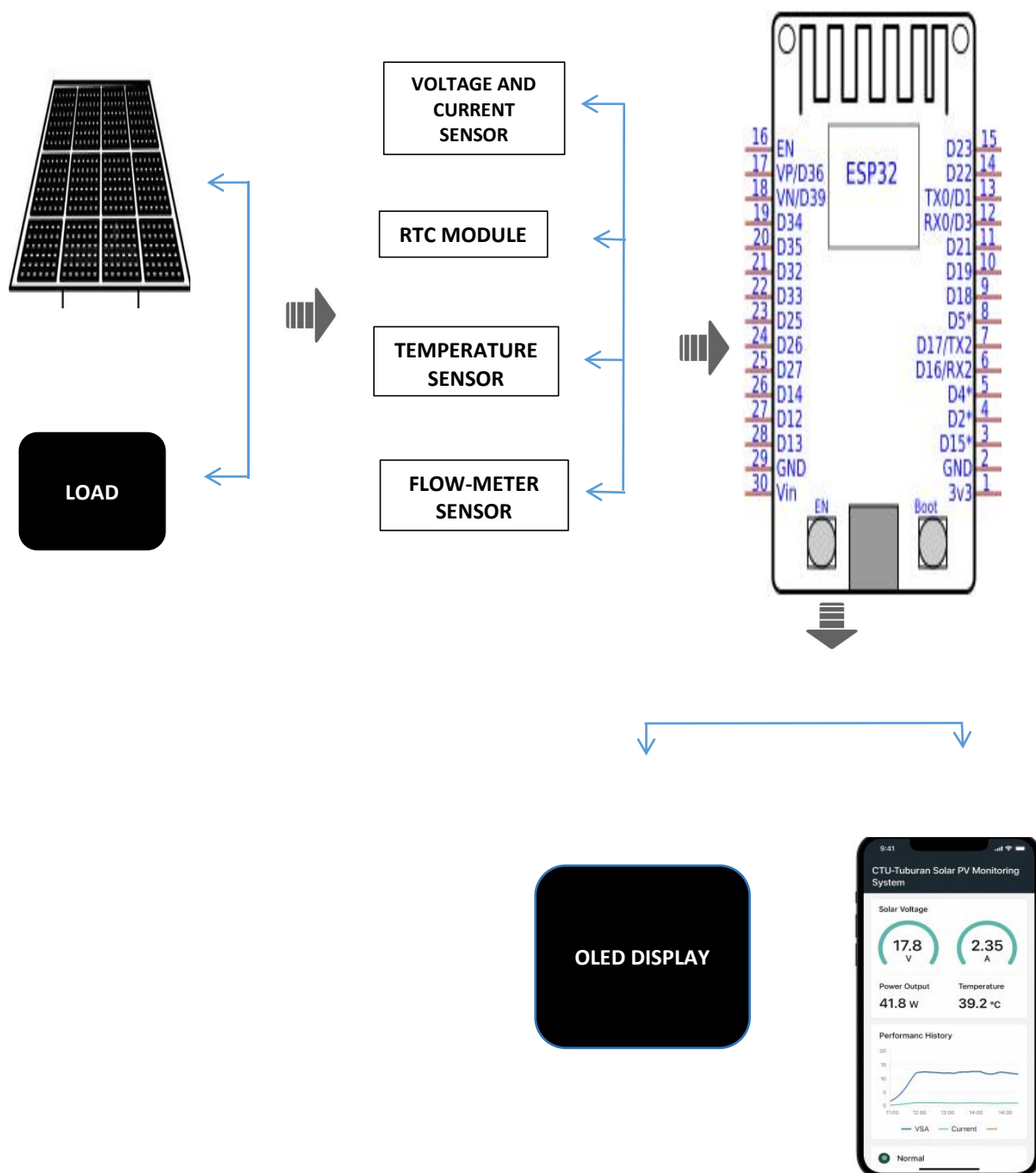
File Edit Sketch Tools Help
Arduino Nano ESP32

sketch.oct11aino
1 #include <Wire.h>
2 #include <Adafruit_INA226.h>
3 #include <Adafruit_GFX.h>
4 #include <Adafruit_SSD1306.h>
5 #include <OneWire.h>
6 #include <DallasTemperature.h>
7 #include <RTClib.h>
8 #include <WiFi.h>
9 #include <FirebaseESP32.h>
10
11 // =====
12 // USER CONFIGURATION
13 // =====
14 #define WIFI_SSID "YOUR_WIFI_SSID"
15 #define WIFI_PASSWORD "YOUR_WIFI_PASSWORD"
16 #define FIREBASE_HOST "https://your-project-id.firebaseio.com/" // Your Firebase URL
17 #define FIREBASE_AUTH "YOUR_FIREBASE_DATABASE_SECRET" // Your Firebase secret key
18
19 // =====
20 // SENSOR PINS & OBJECTS
21 // =====
22 #define ONE_WIRE_BUS 4 // DS18B20 data pin
23 #define SCREEN_WIDTH 128
24 #define SCREEN_HEIGHT 64
25
26 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
27 Adafruit_INA226 ina226;
28 OneWire oneWire(ONE_WIRE_BUS);
29 DallasTemperature sensors(&oneWire);
30
31 // =====
32 // RTC Setup
33 // =====
34 #define RTC_DS3231 rtc;
35
36 // =====
37 // TIMING SETTINGS
38 // =====
39 unsigned long lastSend = 0;
40 const unsigned long sendInterval = 10000; // every 10 seconds
41
42 void setup() {
43   Serial.begin(115200);
44   Wire.begin();
45
46   // --- OLED Setup ---
47   if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
48     Serial.println("OLED not found!");
49     for(;;);
50   }
51   display.clearDisplay();
52   display.setTextSize(1);
53   display.setCursor(0, 0);
54   display.println("Solar PV Monitoring");
55   display.display();
56
57   // --- INA226 Setup ---
58   if (!ina226.begin()) {
59     Serial.println("Failed to find INA226 chip");
60     while(1);
61   }
62   ina226.setCalibration_32V_2A();
63
64   // --- DS18B20 Setup ---
65   sensors.begin();
66
67   // --- RTC Setup ---
68   if (!rtc.begin()) {
69     Serial.println("Couldn't find RTC");
70     while(1);
71   }
72
73   // --- WiFi Setup ---
74   Serial.println("Connecting to WiFi...");
75   WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
76   while (WiFi.status() != WL_CONNECTED) {
77     delay(500);
78     Serial.print(".");
79   }
80   Serial.println("\nConnected to WiFi");
81
82   // --- Firebase Setup ---
83   config.host = FIREBASE_HOST;
84   config.signer.tokens.legacy_token = FIREBASE_AUTH;
85   Firebase.begin(&config, &auth);
86   Firebase.reconnectWiFi(true);
87   Serial.println("Connected to Firebase!");
88
89   // --- Display on OLED ---
90   display.clearDisplay();
91   display.setCursor(0, 0);
92   display.setTextSize(1);
93   display.println("Solar PV Monitor");
94   display.setTextSize(1);
95   display.print("V: %.2f V\n", busVoltage);
96   display.print("I: %.2f A\n", current);
97   display.print("T: %.2f C\n", temperature);
98   display.print("Time: %02d:%02d:%02d\n", now.hour(), now.minute(), now.second());
99   display.display();
100
101   // --- Send Data to Firebase ---
102   if (Firebase.ready()) {
103     String path = "/solar_data";
104     Firebase.setFloat(fdb, path + "/voltage", busVoltage);
105     Firebase.setFloat(fdb, path + "/current", current);
106     Firebase.setFloat(fdb, path + "/temperature", temperature);
107     Firebase.setString(fdb, path + "/time", String(now.timestamp(DateTime::TIMESTAMP_TIME)));
108     Firebase.setString(fdb, path + "/status", "Normal");
109   } else {
110     Serial.println("Firebase not ready!");
111   }
112 }
113
114 void loop() {
115   if (millis() - lastSend >= sendInterval) {
116     lastSend = millis();
117
118     // --- Read Sensors ---
119     float busVoltage = ina226.getBusVoltage_V(); // Voltage in Volts
120     float current = ina226.getCurrent_mA() / 1000.0; // Convert mA to A
121     sensors.requestTemperatures();
122     float temperature = sensors.getTempCbyIndex(0);
123     DateTime now = rtc.now();
124
125     // --- Display on Serial Monitor ---
126     Serial.println("=====");
127     Serial.print("Voltage: %.2f V\n", busVoltage);
128     Serial.print("Current: %.2f A\n", current);
129     Serial.print("Temperature: %.2f C\n", temperature);
130     Serial.print("Time: %02d:%02d:%02d\n", now.hour(), now.minute(), now.second());
131     Serial.println("=====");
132
133     // --- Display on OLED ---
134     display.clearDisplay();
135     display.setCursor(0, 0);
136     display.setTextSize(1);
137     display.println("Solar PV Monitor");
138     display.setTextSize(1);
139     display.print("V: %.2f V\n", busVoltage);
140     display.print("I: %.2f A\n", current);
141     display.print("T: %.2f C\n", temperature);
142     display.print("Time: %02d:%02d:%02d\n", now.hour(), now.minute(), now.second());
143     display.display();
144
145     // --- Send Data to Firebase ---
146     if (Firebase.ready()) {
147       String path = "/solar_data";
148       Firebase.setFloat(fdb, path + "/voltage", busVoltage);
149       Firebase.setFloat(fdb, path + "/current", current);
150       Firebase.setFloat(fdb, path + "/temperature", temperature);
151       Firebase.setString(fdb, path + "/time", String(now.timestamp(DateTime::TIMESTAMP_TIME)));
152       Firebase.setString(fdb, path + "/status", "Normal");
153     } else {
154       Serial.println("Firebase not ready!");
155     }
156   }
157 }

```

Step 4. Product Setup

After the programming and testing phases, the complete system is installed into its final setup location. All components such as the ESP32, INA226, DS3231 RTC module, buck converter, and internal wiring are neatly arranged and securely mounted inside the weatherproof enclosure, while the OLED display is positioned to remain visible from outside the casing for convenient monitoring. The YF-S201 flow meter is installed inline with the existing water pipeline, ensuring that the arrow marking aligns with the direction of water flow and is tightly secured using PVC fittings or hose clamps to prevent leakage. All cables are routed through cable glands and sealed properly to protect against dust, rain, and outdoor exposure. The enclosure is then mounted near the solar water pump system, while the photovoltaic panel remains positioned in an open area with maximum sun exposure for proper power generation. The system is connected to the regulated 5V output from the XL7015 DC-DC converter, which draws power from the charge controller. Once powered on, the OLED display shows initial real-time readings for voltage, current, temperature, water flow rate, and power output. The system then connects to the Wi-Fi network and automatically uploads data to the Firebase Realtime Database. Users can now access the monitoring dashboard via a mobile app or web interface, enabling continuous and remote visualization of solar energy performance and water consumption in the CTU Tuburan campus installation.

Figure 6. Product Set-up

Preliminary Procedure

To ensure the validity, reliability, and functionality of the IoT-Based Monitoring System for Solar Photovoltaic Arrays at Cebu Technological University – Tuburan Campus, a preliminary procedure will be conducted before full-scale implementation.

The researchers first performed a pilot test of the IoT-based monitoring prototype to identify potential technical issues in data transmission, sensor calibration, and system responsiveness. This initial testing verified whether the sensors could accurately measure voltage, current, and temperature while maintaining stable Wi-Fi communication through the ESP32 microcontroller and the dashboard platform.

A pre-survey evaluation of the research questionnaire was also conducted among a small group of respondents consisting of students, maintenance staff, and campus officer. This pilot survey helped determine the clarity, relevance, and appropriateness of the questionnaire items and response options. Feedback obtained from this phase was used to refine the survey instrument, ensuring it was comprehensible and effective for the actual data collection phase.

Additionally, sensor calibration and system validation were performed to guarantee accurate data collection. The INA226 voltage and current sensor, DS18B20 temperature sensor, and OLED display were individually tested for accuracy and consistency.

This preliminary stage ensured that both the IoT system and research instruments were technically sound, user-friendly, and ready for full deployment and evaluation within the university's solar energy setup.

Dry Run Procedure

The dry run for the IoT-Based Monitoring System of the Solar Photovoltaic (PV) installation at Cebu Technological University–Tuburan Campus will be conducted to ensure the accuracy, reliability, and functionality of the system before full implementation. The process will begin with pre-dry run checks where the solar PV panels, sensors, and IoT devices are inspected to confirm proper installation and secure electrical connections. The microcontroller and communication modules will then be powered on, ensuring that all devices are operational and connected to the campus internet network.

Once the system is initialized, the sensors will be tested to collect baseline readings for voltage, current, power output, and temperature, which will then be verified against manual readings using a multimeter to ensure accuracy. Data transmission will also be tested by sending sensor data to the cloud or local server to check for any delays, missing records, or disconnection issues. After this, the monitoring dashboard will be assessed to confirm if it displays real-time solar PV performance data accurately, including graphs, and logs.

To further validate the system's stability, a reliability test will be performed by running the system continuously for two to four hours under varying sunlight conditions. During this period, the performance of the sensors, the consistency of data transmission, and the responsiveness of the dashboard will be closely monitored. Any errors, fluctuations, or downtime will be recorded.

Throughout the dry run, safety precautions will be strictly observed by ensuring that electrical connections are insulated and that direct handling of the PV panels is avoided during peak sunlight hours. All observations, screenshots of dashboard outputs, and photographs of the setup will be documented for evaluation. The expected outcome of the dry run is that the IoT sensors provide accurate data, the system reliably transmits information to the monitoring platform, and the dashboard reflects real-time solar performance. If issues are encountered, corrective actions will be identified and applied before the official deployment of the system.

Data Collection Procedure

A. Clear Data Collection

The researchers submitted a transmittal letter in order to collect and gather data to the Campus Director of Cebu Technological University Tuburan, Campus.

B. Preliminary

After the approval of the letter, the researchers set a schedule to conduct a pilot testing of the modified survey questionnaire to three key respondent groups—students, maintenance staff, and campus officials—who were directly involved in the operation and management of the solar panel system in the field of technology. On the other hand, the questionnaire underwent research instrument validation by the experts.

C. Administration of Questionnaire

In gathering data for the actual data collection, letters of consent Were sent from the Campus director and College of Technology – BIT

Department. After the letters were approved, the researchers set a schedule with the respondents specifically in the data gathering phase. Additionally, the actual data gathering was done using the survey questionnaires where in the respondents could freely answer the questions based from their observations of the product.

Statistical Treatment Data

After retrieving the questionnaires, the frequencies of the responses were collated, tabulated, tallied, and subjected to the following statistical procedures and analysis.

1.Simple Percentage. This statistical tool was utilized to determine the percentage of the respondents.

2.Weighted Mean. This statistical tool was utilized to determine the level of effectiveness and the product's acceptability of an IoT-Based Monitoring System for Solar Photovoltaic Arrays.at Cebu Technological University, Tuburan Campus:

3. Spearman-rho-correlation. This was used to determine the significant relationship between the perception of the respondents on the level of effectiveness and acceptability of the developed product.

Scoring Procedure

In the scoring procedure, as shown below, the gathered data was interpreted through the use of the Likert five-point scale, wherein the respondents answered on the scale, range, category, and its verbal description.

Table 4. Scoring of Product Effectiveness**Part 1.Product Effectiveness of Solar Photovoltaic Arrays Monitoring****System**

Scale	Range	Verbal Description		Interpretation
5	4.21 – 5.00	Highly (HE)	Effective	The system is excellent in performance, reliability, and functionality. It fully achieves its intended purpose with outstanding results.
4	3.41 – 4.20	Effective (HE)		The system performs well and meets user expectations in most aspects of operation and monitoring.
3	2.61 – 3.40	Moderately Effective (ME)		The system functions adequately but requires improvements to enhance performance or stability.
2	1.81 – 2.60	Ineffective (IE)		The system demonstrates low effectiveness and requires significant enhancements to achieve reliable performance.
1	1.00 – 1.80	Highly (HE)	Effective	The system fails to perform its intended purpose and does not meet user or operational expectations.

Table 5. Scoring of Product Acceptability

Part 2. Product Acceptability of Solar Photovoltaic Arrays Monitoring System

Scale	Range	Verbal Description	Interpretation
5	4.21 – 5.00	Highly Acceptable (HA)	The product is excellent in quality, design, and overall user satisfaction. It fully meets or exceeds user expectations.
4	3.41 – 4.20	Acceptable (A)	The product performs well and satisfies most users' needs, with only minor improvements needed.
3	2.61 – 3.40	Moderately Acceptable (MA)	The product is acceptable but needs improvement in certain features or usability aspects.
2	1.81 – 2.60	Unacceptable (UA)	The product shows low acceptability and requires major revisions to meet user needs.
1	1.00 – 1.80	Highly Unacceptable (HA)	The product is poor in quality, design, or performance and fails to meet user expectations.

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APPENDICES

APPENDIX A
TRANSMITAL LETTER

APPENDIX B

SURVEY QUESTIONNAIRE

Assessing the Effectiveness and Acceptability of Photovoltaic Array Monitoring System at Cebu Technological University - Tuburan Campus

Part I. The level of effectiveness of a Photovoltaic Array Monitoring System

Directions: Put a checkmark (/) in the column below that corresponds to the statement.

Legend:

5- Highly Effective (HE)

4-Effective (E)

3-Moderately Effective (ME)

2-Ineffective (I)

1- Highly Ineffective (HE)

ASPECT	5 (HE)	4 (E)	3 (ME)	2 (I)	1 (HE)
A. Performance					
The system efficiently collects real-time data from the solar panels.					
The monitoring system consistently transmits accurate data without delay.					
The performance remains stable even during fluctuating environmental conditions.					

The automated monitoring enhances the reliability and stability of solar power generation in distributed PV systems. The monitoring tools are effective in maximizing energy yield and reducing downtime in solar photovoltaic installations					
B. Functions	5	4	3	2	1
The IoT system functions effectively in tracking voltage, current, and energy output The system provides automated alerts or notifications for maintenance issues. The system performs its intended function with minimal human supervision The device determines the level of water.					
C. Convenience	5	4	3	2	1
Users can easily access the monitoring data remotely through the dashboard. The system simplifies solar panel management and reduces manual inspection time. The interface allows quick updates and monitoring with minimal training					

D. Serviceability	5	4	3	2	1
The IoT components are easy to maintain and replace when necessary. Troubleshooting and diagnostics can be performed easily within the system The dashboard includes indicators that help in identifying system faults Maintenance procedures are clearly outlined and supported by the system interface.					
E. Perceived Quality	5	4	3	2	1
The overall quality of the system meets the expectations for a reliable solar monitoring tool. The IoT monitoring system ensures dependable performance over time The device and dashboard are made from high-quality, long-lasting materials The system provides accurate and trustworthy data for decision-making and maintenance planning.					

Part II. The product acceptability of a smart multipurpose lift table with integrated safety and IoT features

Directions: Put a checkmark (/) in the column below that corresponds to the statement.

Legend:

Highly Acceptable (HA)

Acceptable (A)

Moderately Acceptable (MA)

Unacceptable (U)

Highly Unacceptable (HU)

ASPECT	5 (HA)	4 (A)	3 (MA)	2 (U)	1 (HU)
A. Perceived Usefulness					
The IoT monitoring system improves the efficiency of solar energy tracking					
The system helps detect faults early, reducing energy loss and maintenance costs.					
Real-time monitoring enhances awareness of solar panel performance					
The system supports sustainable campus energy management					
The product contributes to better decision-making in energy conservation.					

B. Perceived Ease of Use	5	4	3	2	1
The IoT monitoring dashboard is simple to understand and operate It is easy to navigate the local LCD and remote monitoring interface. Learning to use the system requires minimal training. The display and controls are accessible to both technical and non-technical users. The user interface is well-designed and intuitive.					
C. Behavior Intention of Use	5	4	3	2	1
Users intend to regularly use the monitoring system for checking solar performance Maintenance staff plan to rely on the system during disasters or when away from campus. The IoT dashboard will be used for continuous monitoring and reporting. Users are likely to recommend the system for other campus installations The system will be integrated into the university's routine maintenance process.					

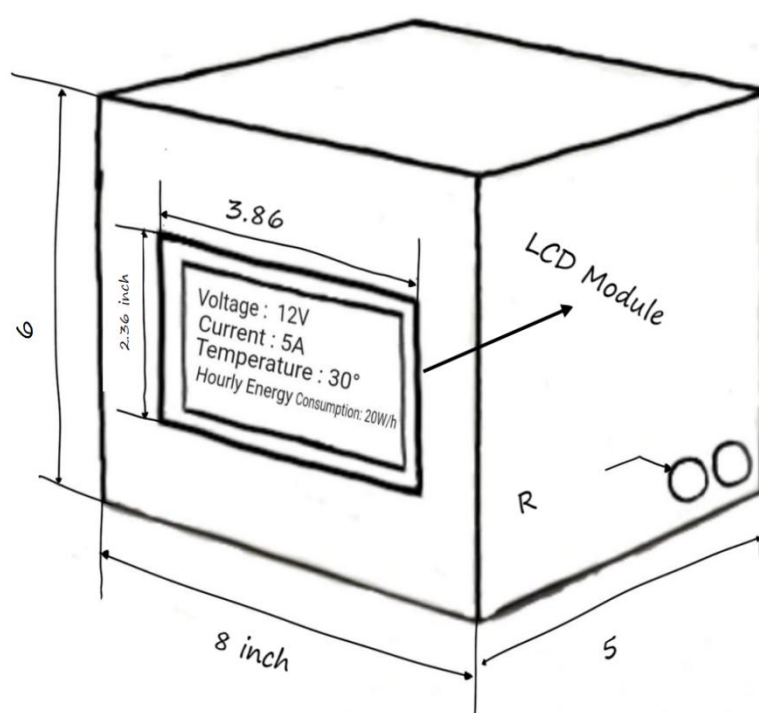
D. Actual Usage	5	4	3	2	1
The system is being used effectively for real-time monitoring of solar performance. Users interact with the system regularly to assess energy generation and performance. The IoT dashboard is accessed during emergencies or equipment failures. The data collected from the system is used to improve maintenance schedules. The monitoring system has become part of daily operations for the solar facility.					

Part 3: Suggestions/Recommendations

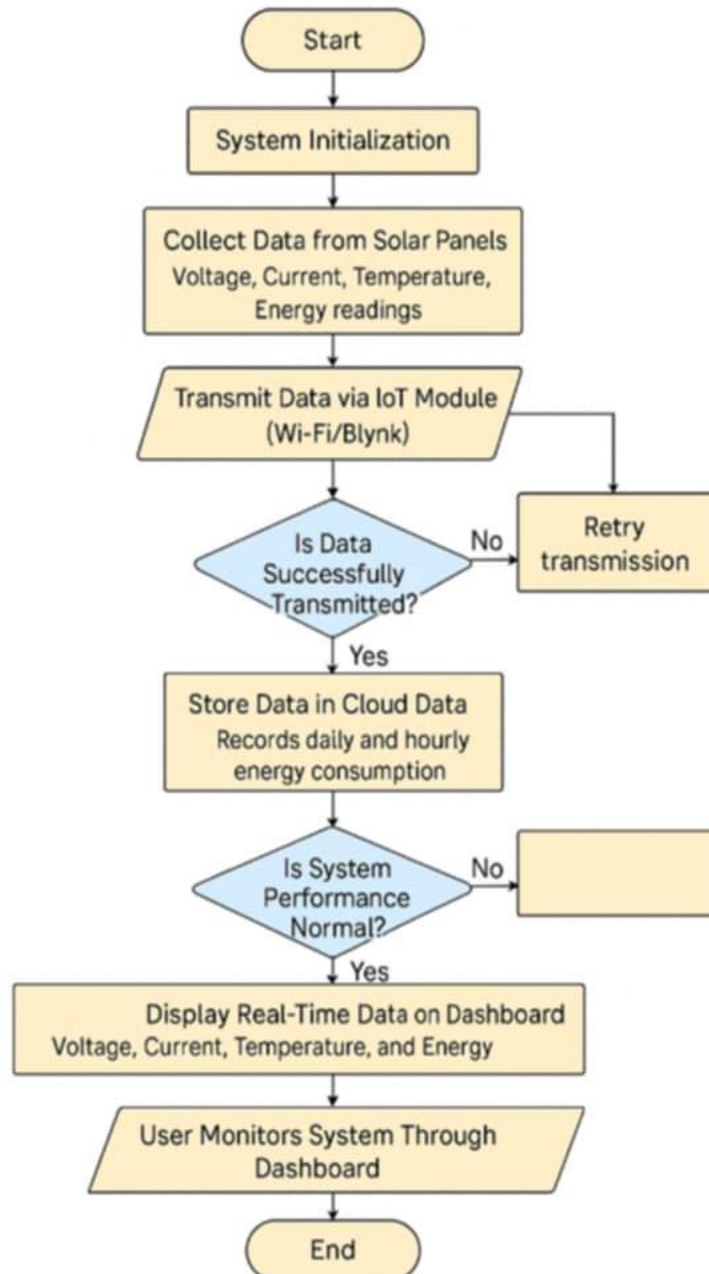
Instructions: Please give some suggestions/recommendations regarding our conducted study

APPENDIX C

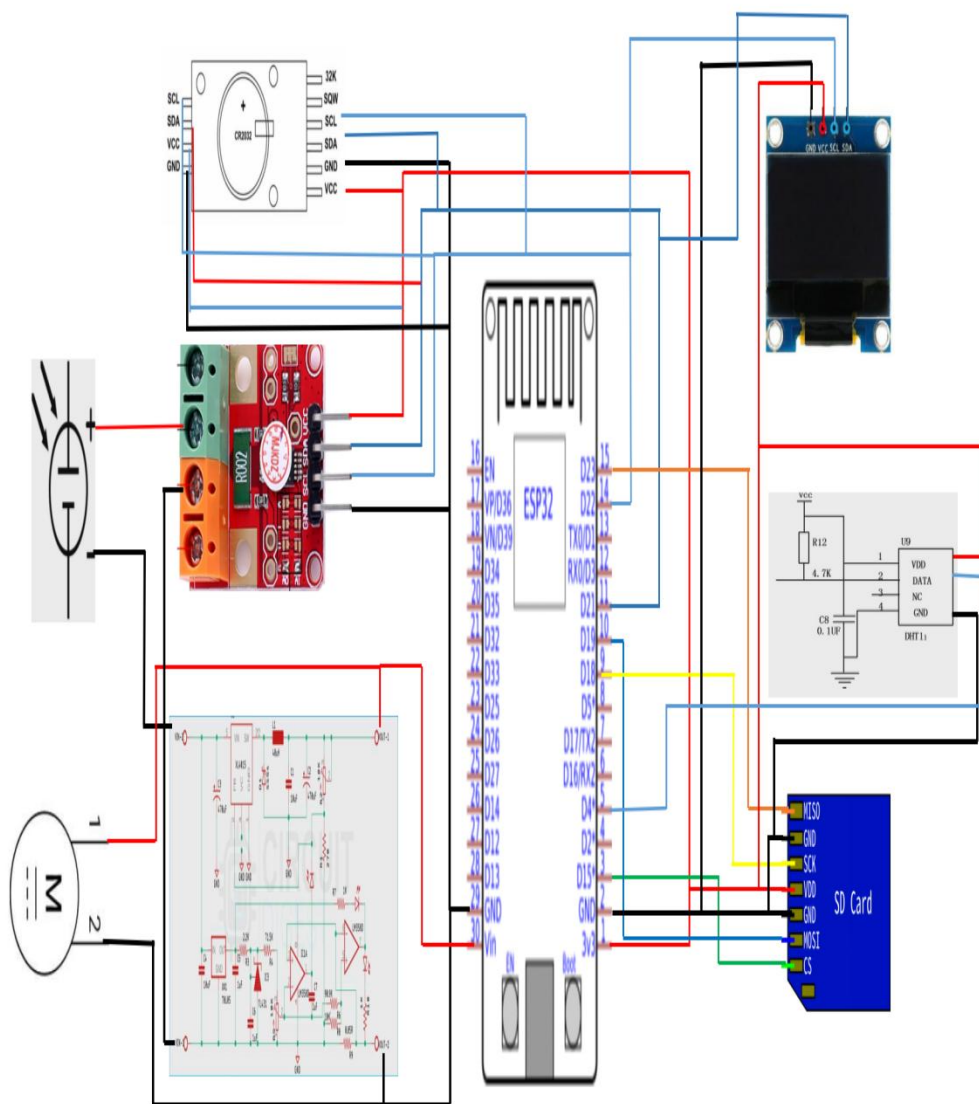
C.1 PRODUCT DESIGN



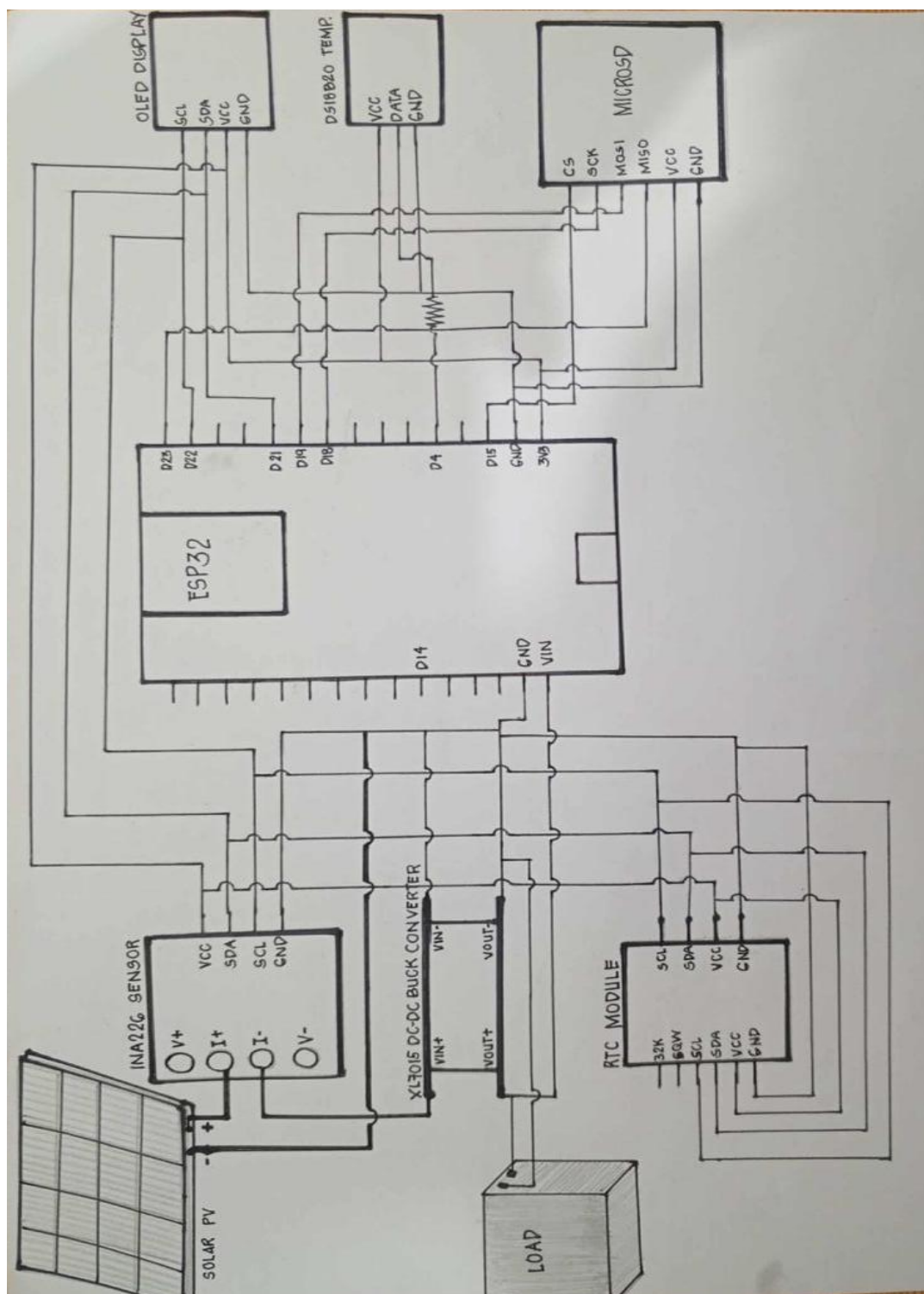
C.2 FLOW DIAGRAM



C.3 SCHEMATIC DIAGRAM



C.4 BLOCK DIAGRAM



APPENDIX D

Estimated Project Costing

Item/s	Quantity	Unit Price	Amount
A. Electrical and Electronics Peripheral			
ESP32 (30 pins)	1	179	179
INA226 Voltage and Current Sensor	1	972	972
DS18B20 Temperature Sensor	1	124	124
XL7015 DC-DC Buck Converter	1	40	40
YF-S201 Flow Meter Sensor	1	146	146
OLED Display (1.3" I ² C)	1 set	139	139
Resistors (4.7k Ω , 10k Ω)	2 set	30	60
Set Jumper Wires	1	209	209
PCB or Breadboard	1	212	212
Weatherproof Enclosure (IP65 or IP67)	1	247	247
Several Cable Glands and Bolts	1	310	310
Sub Total:			2638.00
Item/s	Quantity	Unit Price	Amount
B. Accessories			
Soldering iron + solder wire	1 set	125	125
Wire stripper and cutter	1	97	97
Multimeter	1	142	142
Small screwdriver set	1 set	91	91

Drill with hole saw bits	1 set	119	119
Electrical tape	1	20	20
Epoxy	1	87	87
USB cable (ESP32 compatible)	1	68	68
Silicone sealant	1	80	80
Enclosure Box	1	255	255
PVC Fittings for Flow Meter Installation	1	92	92
Teflon tape / plumbing sealant	1	40	40
Hose clamps	1	19	19
			Sub Total: 1235.00

C. Material Estimated Cost 3873.00

D. Total Cost 3873.00

APPENDIX E

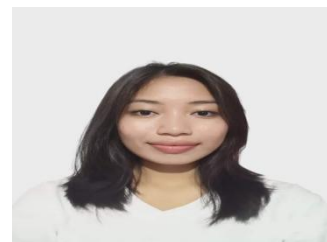
CURRICULUM VITAE

RECHELL ANN SUNGKIP

Ubogon, Asturias, Cebu

09265353973

sungkiprechellann@gmail.com



PERSONAL DATA

Date of Birth :	July 11, 2005
Gender :	Female
Nationality :	Filipino
Height :	5'2
Weight :	55 kg
Civil Status :	Single
Religion :	Roman Catholic
Place of Birth :	Ubogon, Asturias, Cebu
Father's Name :	Ranel B. Sungkip
Mother's Name :	Elsa P. Sungkip

EDUCATIONAL BACKGROUND

College :	Cebu Technological University -Tuburan Campus Barangay VIII, Tuburan, Cebu Bachelor of Industrial Technology Major in Computer Technology October 2025-Present
Senior High :	Nangka National High School Nangka, Balamban, Cebu July 2022
Junior High :	Looc Norte National High School Looc Norte, Asturias, Cebu April 2019
Elementary :	Ubogon Elementary School Ubogon, Asturias, Cebu March 2017

WORK EXPERIENCE OR TRAINING ATTENDED

- Coolway Multipurpose Cooperative
Aliwanay, Balamban, Cebu
December 2021



+639702107149
suelesarci@gmail.com

PROFESSIONAL SUMMARY

Obtain a challenging leadership position that will utilize my lean management skills and benefit from my organizational abilities, positive attitude and strong customer service skills.

EDUCATION

- Asian College – Quezon City Bookkeeping NC3 (2019)

SKILLS

- Accurate money handling
- Inventory management
- Time management skills
- Multi-tasking ability
- Adaptability
- Computer proficiency
- Creative problem solver
- Quick learner
- Tech-savvy

SOFTWARE & TOOL

- Microsoft Office
- G-Suite
- SAP Software System
- Canva
- Lightroom
- J E Software System
- Focus ERP System
- Outlook
- Power Point
- Adobe Suite

RESUME

Hello, Everyone

ARCILLIN E. SUELES

PURCHASING | Accounting COORDINATOR

WORK EXPERIENCE

PURCHASING COORDINATOR | Accounts Payable | US Account

YFC-Boneagle INTERNATIONAL INC.

May 2019 - OCTOBER 2021

- Keeping track of all the payments and expenditures, including purchase orders, invoices, statements, etc.
- Reconciling processed work by verifying entries and comparing system reports to balances.
- Maintaining historical records.
- Paying vendors by scheduling pay checks and ensuring payment is received for outstanding credit ; generally responding to all vendor inquiries regarding finance
- Preparing analysis accounts and producing monthly reports
- Helps assists for checking order payment documentary
- Continuing to improve the payment process

Tax Associate | Accounts Receivable | US Account

- Ensuring clients to comply the IRS regulations and meet their state federal tax obligations.
- Researching tax certificate issues and assisting tax audits.
- Consulting with clients, reviewing US tax information certificates and responding to questions and concern.
- Helps assist in gathering documentation of tax jurisdiction audits, tax refunds, tax adjustments, notation, online billing, etc.
- Maintaining accurate and up-to-date client records.

Junior Tax Analyst

Gramono Properties and Development Corporation

June 2017 – May 2019

- Produced monthly reports using advanced Excel spreadsheet functions.
- Organized forms, made photocopies, filed records and prepared correspondence reports.
- Confirm that entered data accurately align with original documentation
- Organized and maintain original paper evidence.
- Assure files are properly prepared and saved to back-up files.
- Organized files in logical and manageable fashion.
- Perform any other tasks that management requires assistance with (sending emails, answering phones, etc.)
- Files tax documents by preparing properly tax returns, municipal returns, extensions and quarterly payments. Like monthly and quarterly Vat, Expanded, Compensation, Map, Swat, Vat Relief)
- Encoding Vat Relief (BIR Transaction Every Quarterly)
- Preparing other tax document.
- Monitoring and checking the SSS, Pag-ibig & Philhealth Rem. & Loan

Charline Aliviado
Bairan,Asturias, Cebu
09463399495
charlinealiviado2@gmail.com



PERSONAL DATA:

AGE : 21 year's old
GENDER : Female
NATIONALITY : Filipino
BIRTH DATE : Sept,09,2004
BIRTH PLACE : Bairan,Asturias, Cebu
CIVIL STATUS : Single
RELIGION : Born again
HEIGHT : 5'0
WEIGHT : 44kg.
FATHER'S NAME : Elorde Aliviado
MOTHER'S NAME : Adelaida Aliviado

EDUCATIONAL BACKGROUND

COLLEGE	Cebu Technological University BRGY.8 Tuburan Cebu Bachelor of Industrial Technology Major in Computer Technology :2025-Present
Senior High	Nicolas U. Tiongko Memorial National High School Dansan, Danao, City July 2022
Junior High	Nicolas U. Tiongko Memorial National High School Dansan, Danao, City April 2018
Elementary	Danasan Elementary School Dansan, Danao, City March 2016

MIRAFLORE B. CASTRO

BRGY COGON, TUBURAN, CEBU

09942110139

miraflorcastro@gmail.com**PERSONAL DATA:**

AGE : 21 years old
 GENDER : Female
 NATIONALITY : Filipino
 BIRTH DATE : September 8, 2004
 BIRTH PLACE : COGON, TUBURAN, CEBU
 CIVIL STATUS : Single
 RELIGION : Roman Catholic
 HEIGHT : 5'3
 WEIGHT : 100 lbs.
 Father's Name : Alijandro Castro
 Mother's Name : Elena Castro

EDUCATIONAL BACKGROUND

COLLEGE	Cebu Technological University - Tuburan Campus Cebu Brgy.8 Tuburan Cebu Bachelor of Industrial Technology major in Computer Technology 2025-Present
Senior High	Tuburan National High School Barangay 7 Tuburan Cebu July 2022
Junior High	: Tuburan National High School Barangay 7 Tuburan Cebu April 2020
Elementary	: Tuburan Central School Barangay 5 Tuburan Cebu Cebu March 2016

WORK EXPERIENCE OR TRAINING ATTENDED

Coast Guard Training Participant – Grade 12 Student

Participated in a training program conducted by the Philippine Coast Guard, gaining basic knowledge of maritime safety, discipline, and teamwork.

JEANE MARIE OLASIMAN

DAAN LUNGSOD TUBURAN, CEBU

09073648030

olasimanjeanemarie@gmail.com**PERSONAL DATA:**

AGE : 21 years old
 GENDER : Female
 NATIONALITY : Filipino
 BIRTH DATE : JULY 7, 2004
 BIRTH PLACE : DAAN LUNGSOD TUBURAN, CEBU
 CIVIL STATUS : Single
 RELIGION : Roman Catholic
 HEIGHT : 5'3
 WEIGHT : 100 lbs.
 Father's Name : Josephene Olasiman
 Mother's Name : Julieta Olasiman

EDUCATIONAL BACKGROUND

COLLEGE : Cebu Technological University -Tuburan
 Campus Cebu

Brgy. 8 tuburan Cebu
 Cebu Bachelor of Industrial Technology major in Computer Technology
 2025-Present

Senior High : Montealegre National High School
 Montealegre Tuburan Cebu
 July 2022

Junior High : Montealegre National High School
 Montealegre Tuburan Cebu
 April 2020

Elementary : Siotes Elementary School
 Siotes Tuburan Cebu
 Cebu March 2016

WORK EXPERIENCE OR TRAINING ATTENDED

ROTC officer
 Cebu Technological University – Tuburan Campus ROTC Unit
 2023-2024

Name: KHENDERLYN A. COMCOM
Contact No.: 09945231282
Email: khenderlyncomcom9@gmail.com
Address: Nahus-an, Sogod, Cebu



I. PERSONAL INFORMATION

Date of birth : February 2, 2005
Age : 20
Sex : F
Civil Status. : Single
Nationality. : Filipino
Religion. : Roman Catholic

II. EDUCATIONAL BACKGROUND

Tertiary Education : Cebu Technological University - Tuburan
Campus
Bachelor of Industrial Technology - Major in
Computer Technology
Brgy. 8, Tuburan, Cebu
August 2023 – Present

Secondary Education : Kalangahan National High School
Brgy. Kalangahan, Tuburan, Cebu
July 2023

Elementary Education : Nahus- an, Elementary, School
Nahus- an, Sogod, Cebu.
April 2016

QUEENSEA ANN G. BUHIA
SAN ROQUE, ASTURIAS, CEBU
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queenseaannbuhia@gmail.com



PERSONAL INFORMATION

Age:21

Gender:Female

Nationality:Filipino

Birthdate:September 10,2004

Birthplace: San Roque, Asturias, Cebu

Religion:Roman Catholic

Height:4'8

Weight: 35.8

EDUCATIONAL BACKGROUND

• ELEMENTARY SCHOOL

San Roque Elementary School S. Y 2015-2016

• JUNIOR HIGH SCHOOL

Looc Norte National High School S. Y 2020-2021

•SENIOR HIGH SCHOOL

Nangka National High School S. Y 2021-2022

•COLLEGE

Cebu Technological University present -2025

Justine O. Ynoy
 BRGY 8, TUBURAN, CEBU
09608522182
 Justinynoy012@gmail.com



PERSONAL DATA:

AGE	: 23 years old
GENDER	: Male
NATIONALITY	: Filipino
BIRTH DATE	: Nov. 27, 2002
BIRTH PLACE	: Old Bago, Asturias, CEBU
CIVIL STATUS	: Single
RELIGION	: Roman Catholic
HEIGHT	: 5'7
WEIGHT	: 58 kg.
Father's Name	: Artemio Ynoy
Mother's Name	: Amor Oppus

EDUCATIONAL BACKGROUND

COLLEGE	: Cebu Technological University -Tuburan Campus Cebu : BRGY. 8 tuburan Cebu :Cebu Bachelor of Industrial Technology major in Computer Technology :2025-Present
Senior High	: Tuburan National High School : BRGY. 7, Tuburan, Cebu : July 2022
Junior High	: Tuburan National High School : BRGY. 7, Tuburan ,Cebu : April 2018
Elementary	: Lunas II Elementary School : Lunas, Asturias, Cebu : Cebu March 2016

WORK EXPERIENCE OR TRAINING ATTENDED

Bureau of Jail Management and Penology